

Humaira Ali

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Education

Bachelor of Technology in Electronics and Communication Engineering 2023 – 2027 (Expected)
VIT Bhopal University, Sehore, Madhya Pradesh
CGPA: **8.62 / 10**
10th: **89.8/100 Carmel Convent BHEL-CBSE (2021)**
12th: **80.4/100 Carmel Convent BHEL-CBSE (2023)**

Technical Skills

Programming: C, C++ (Embedded Systems), Java, Python (Basic)

Embedded Systems: ESP32, Arduino, Embedded C, Firmware Development, Sensor Interfacing, Hardware–Software Integration

Communication: Bluetooth, RFID, UART/Serial Communication, Wireless Communication

Core Engineering: Digital Electronics, Microprocessors & Microcontrollers, Digital Signal Processing, Communication Systems, Control Systems

Tools: Altium Designer, MATLAB, Keil uVision, LTspice, Proteus, SQLite

Projects

Smart Helmet System for Accident Detection & Navigation

- Developed an ESP32-based embedded safety system as a multi-sensor real-time firmware architecture integrating MPU6050 and ultrasonic sensors for crash detection, rear-obstacle detection, and rider assistance.
- Implemented a 3G threshold-based accident detection algorithm with a 10-second cancellation window, Bluetooth communication, and GPS-enabled SOS alerts.
- *Tech:* ESP32, Embedded C++, Bluetooth, MPU6050, Ultrasonic Sensor, Android

RFID-Based Multi-Module Campus Management System

- Developed an ESP32- and RFID-based campus management platform integrating attendance, library management, and cafeteria billing across 3 operational modules.
- Automated attendance logging, PDF bill generation, Excel exports, and database management through a Python GUI application.
- *Tech:* ESP32, Python, SQLite, RFID, Bluetooth, Tkinter

Morse Code Encoder–Decoder Device

- Designed a bidirectional Arduino-based Morse code encoder and decoder supporting configurable transmission speeds from 100–600 ms.
- Implemented dual communication modes, LCD-based decoding, adaptive timing control, SOS signaling, and configurable encoding logic.
- *Tech:* Arduino, Embedded C++, LCD, Signal Encoding

Certifications

- Introduction to Internet of Things – NPTEL
- Introduction to Programming in C – NPTEL
- MATLAB Signal Processing Certifications – MathWorks
- VLSI Design & Verification Program – Ethnus
- Bits and Bytes - Coursera
- Google IT support - Coursera
- Introduction To self Driving Cars - Coursera

Experience

Vocational Training in BHEL (6-weeks)

- Project on Programmable Logic Controller (PLC) Based Automation Systems at BHARAT HEAVY ELECTRICALS LIMITED (BHEL) BHOPAL.